SMART@Home Hackathon Brief

The clinician data view problem. Background for hackathon participants and their AI coding assistants.

Cincinnati Children's Hospital Medical Center

1. What SMART@Home is

SMART@Home (Self-Management Assistance for Recommended Treatment at Home) is a software platform built at Cincinnati Children's. A care team and its patients use it to create a home self-management app together, put it on the patient's phone the same day, and keep changing it as they learn what works. No programmer or app store release is needed for a change.

The patient sees one app. Inside that app are one or more **trackers**. A tracker is the set of questions, medication schedules, reminders, goals, education and device readings for one condition. A patient with asthma and migraine, for example, can have an asthma tracker and a migraine tracker in the same app. A straightforward tracker takes about 10 to 15 minutes to set up.

Asthma is the first condition in clinical use. A randomized pilot study of the asthma tracker in adolescents aged 12 to 18 is open and recruiting at Cincinnati Children's (ClinicalTrials.gov NCT06159127).

2. How the data are collected

Patients, usually adolescents, enter data at home on an iPhone app. An Android version is in development. The app collects:

- **Answers to tracker questions.** Question types include yes or no, numbers, face scales, icon choices, checkboxes, single-choice lists and 1 to 10 scales. Examples: "How are you feeling today?" and "Which of these symptoms do you have right now?"
- **Medication doses.** The app reminds the patient to take each medication and records when the patient marks a dose as taken. This is the patient's own report, not a verified dose.
- **Home spirometry.** A spirometer measures how much air a person can blow out and how fast. SMART@Home pairs over Bluetooth with the MIR Spirobank Smart home spirometer. Readings include FEV1 (the volume of air blown out in the first second), FVC (the total volume blown out), their ratio, and peak flow (the fastest rate of air flow). The device checks each blow against American Thoracic Society and European Respiratory Society quality standards and coaches the patient, for example "keep going until completely empty."
- **Asthma action plan status.** The patient's own written asthma action plan is built into the app. Using symptoms and spirometry, the app tells the family which step of that plan applies (a green, yellow or red zone).

- **Rewards.** Badges and streaks that reward regular use.

Entries are time-stamped and sent to a secure server in near real time. Care team staff see them in a web portal, where they also register patients and assign trackers. At Cincinnati Children's, clinicians can also open a SMART@Home tab inside the patient's Epic chart. Data flow one way, from the app to the clinician view. They are not written into the medical record.

3. How the data are stored

This section is mainly for AI coding assistants. Each patient has one nested JSON document (stored in Azure Cosmos DB for MongoDB). The main parts are:

- `dailyLogs[]`, one entry per day, each holding `trackerResponses[]` and `medicationAdministrations[]`
- `responses[]` inside each tracker response, holding a `questionId` and a `value`
- `medications[]`, `reminders[]`, `medicalDevices[]` with `baselineReadings[]` and quality reports
- `asthmaTreatmentPlan` (status, expiration, retest time)
- `earnedMedicationBadges[]` and `earnedTrackerBadges[]`

Things that make the data hard to use directly:

- A patient can submit a tracker many times on the same day. Entries arrive at irregular times.
- Many fields are optional and often empty.
- The `value` field is always text, even when it holds a number, a list of symptoms (written as a JSON list) or a device reading. It has to be read according to the question type.
- The question wording lives in the tracker definition, not in the patient's record. Medications are linked to doses by long random ID codes.

Use made-up or de-identified example data when building. Do not put real patient data into an AI tool.

4. What clinicians told us they need

A human factors firm, POMIET, interviewed clinicians in two rounds for this project: six clinicians in May 2022 and four clinicians in the asthma study in August and September 2023. They reported that clinicians:

- Are very busy, see 5 to 30 patients a day, mostly with chronic conditions, and do most of their work in Epic.
- Would most likely look at home data right before a visit, either themselves or through a nurse practitioner.
- Find it hard to compare a patient's answers over time.

- Want a summary, such as a dashboard with graphs and percentages, rather than raw entries.
- Want to know whether the patient is taking the medication, so they can tell a treatment that is not working from a treatment that is not being taken.
- Ask themselves questions such as: How is the disease going? How often are symptoms happening, and how often is treatment used? How many school days were missed? Is the patient following the plan? Is the patient taking the medication?
- Want to use the data with the patient during the visit, for example to show how many days that month were hard.
- In some cases, want to download the data to analyze it themselves.

Different specialists need different displays of the same data. A lung specialist, a gastroenterologist, a neurologist, a primary care physician and a nurse will each look for different things.

5. The problem in two screenshots

The team put its effort into making the patient app work well and making sure the data reach the portal. That part works. Patients use the app, and in several cases patients have asked to keep using it after a study ended. Where the system falls short is the clinician's side. The data arrive, but they are shown in a way a busy clinician cannot use.

Both screenshots below come from the same example account. The data are for illustration only.

Screenshot 1. The full patient record as a spreadsheet

This is everything stored for one patient, laid out as a spreadsheet. It is shown on the next page. It contains six stacked tables, one after another, each with its own column headings:

- **Daily logs, medication administrations:** each dose the patient marked as taken
- **Medications:** the patient's medication list, with schedule and the condition each is for
- **Medical devices:** the paired spirometer
- **Baseline readings:** the individual spirometer blows
- **Quality report:** the device's quality check and coaching message for each blow
- **Earned tracker badges**

Why it is hard to work with:

- **Nothing answers a clinical question.** There is no summary, no trend and no picture of whether lung function or symptoms are changing.
- **Column headings are cut off.** About fifteen spirometry columns all read "FVC Plus R...", so a reader cannot tell which number is FEV1, FVC or peak flow. Only the last column shows a unit (liters). Values run to five or six decimal places.
- **Medications are linked by codes.** Doses point to medications by long random IDs such as `1027ba72...`. "Excedrin" appears twice in the medication list, one archived and

one active. Telling which one a dose belongs to means matching codes by hand.

- **Fields have no clear meaning.** The "All Medications Administered" column reads TRUE for the 2025 doses and FALSE for the 2026 doses, even though a dose is recorded on each row. The sheet does not explain the difference.
- **Device results conflict.** Spirometer blows alternate between Accepted TRUE and FALSE, yet the quality column shows "NotAccept..." on every row. The quality table also has seven rows for six blows, because one blow has two coaching messages.
- **Programmer labels.** Coaching messages appear as run-together text such as "KeepGoingUntilCompletelyEmpty". Columns such as Protocol, Boot Id, Device Type and the device's Bluetooth address mean nothing to a clinician.

DAILY LOGS — MEDICATION ADMINISTRATIONS

Day	All Medications Administered	ID	Time	Name	Value
2025-03-28	TRUE	1027ba72-	20:21	Excedrin	1
2025-03-28	TRUE	1027ba72-	19:21	Excedrin	1
2025-03-28	TRUE	1027ba72-	20:21	Excedrin	1
2026-03-11	FALSE	4aec1b37-	12:47	Dulera 2	1
2026-05-05	FALSE	aecc069e-	19:51	Excedrin	1
2026-05-05	FALSE	4aec1b37-	19:51	Dulera 2	1
2026-06-24	FALSE	4aec1b37-	09:31	Dulera 2	1
2026-06-24	FALSE	aecc069e-	09:31	Excedrin	1

MEDICATIONS

ID	Dosage / Timing / Period Unit	Dosage / T	Dosage / T	Dosage / T	Dosage / T	Dosage / A	Dosage / S	How Often	Icon	Name	Related Co	Archived	Index
4aec1b37-c83a-412a-a4f6-f3d8332c740f	Daily	1	5					Daily	Capsule	Dulera 2	Asthma Ec	FALSE	2
1027ba72-fe17-432d-9904-e72a721d8c2f					TRUE			As Needed	Capsule	Excedrin		TRUE	0
aecc069e-60a2-413d-b596-25054e662cbb					TRUE			As Needed	Capsule	Excedrin	Cerebral P	FALSE	1

MEDICAL DEVICES

Protocol	Address	Boot Id	Serial Nun	Device Typ	Ats Standa	Name	Advertiser	Baseline Reading Date
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-24

MEDICAL DEVICES — BASELINE READINGS

Protocol	Address	Boot Id	Serial Nun	Device Typ	Ats Standa	Name	Advertiser	Baseline R	Accepted	Quality Re	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	Results / FVC (L)
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16879	23.36432	47.93997	4.171884	5.39	5.376757	0	100.2463	2	9.544466	37.1	2.23	1.37	5.39
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	16.23978	28.52428	4.171884	4.68	5.376757	0	101.3622	1.19	9.544466	21.8	1.55	0.76	5.45
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16815	22.52614	41.22837	4.171884	5.19	5.376757	0	101.9202	1.72	9.544466	31.4	2.15	0.89	5.48
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	27.24091	43.62537	4.171884	5.38	5.376757	0	103.4081	1.82	9.544466	32.7	2.6	0.98	5.56
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16879	102.4678	109.7825	4.171884	4.93	5.376757	0	91.69095	4.58	9.544466	92.9	9.78	6.86	4.93
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	113.7832	108.584	4.171884	4.96	5.376757	0	92.24891	4.53	9.544466	91.3	10.86	6.59	4.96

MEDICAL DEVICES — BASELINE READINGS — QUALITY REPORT — QUALITY INDICATIONS

Protocol	Address	Boot Id	Serial Nun	Device Typ	Ats Standa	Name	Advertiser	Baseline R	Accepted	Quality Re	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	FVC Plus R	Quality M	Instructions
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16879	23.36432	47.93997	4.171884	5.39	5.376757	0	100.2463	2	9.544466	37.1	2.23	1.37	5.39	BlowOutLa	KeepGoingUntilCompletelyEmpty
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	16.23978	28.52428	4.171884	4.68	5.376757	0	101.3622	1.19	9.544466	21.8	1.55	0.76	5.45	BlowOutLa	KeepGoingUntilCompletelyEmpty
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16815	22.52614	41.22837	4.171884	5.19	5.376757	0	101.9202	1.72	9.544466	31.4	2.15	0.89	5.48	BlowOutLa	KeepGoingUntilCompletelyEmpty
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16815	22.52614	41.22837	4.171884	5.19	5.376757	0	101.9202	1.72	9.544466	31.4	2.15	0.89	5.48	BlowOutFa	BlastOutImmediatelyWhenCompletelyFull
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	27.24091	43.62537	4.171884	5.38	5.376757	0	103.4081	1.82	9.544466	32.7	2.6	0.98	5.56	BlowOutLa	KeepGoingUntilCompletelyEmpty
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	FALSE	NotAccept	16879	102.4678	109.7825	4.171884	4.93	5.376757	0	91.69095	4.58	9.544466	92.9	9.78	6.86	4.93	BlowOutLa	KeepGoingUntilCompletelyEmpty
9	AAF17BF4-FB64-B5AB-D7C6-27344E9A81FA	SM	Z125775	0		Spirobank	SM-009-Z1	2026-06-2	TRUE	NotAccept	16879	113.7832	108.584	4.171884	4.96	5.376757	0	92.24891	4.53	9.544466	91.3	10.86	6.59	4.96	BlowOutLa	KeepGoingUntilCompletelyEmpty

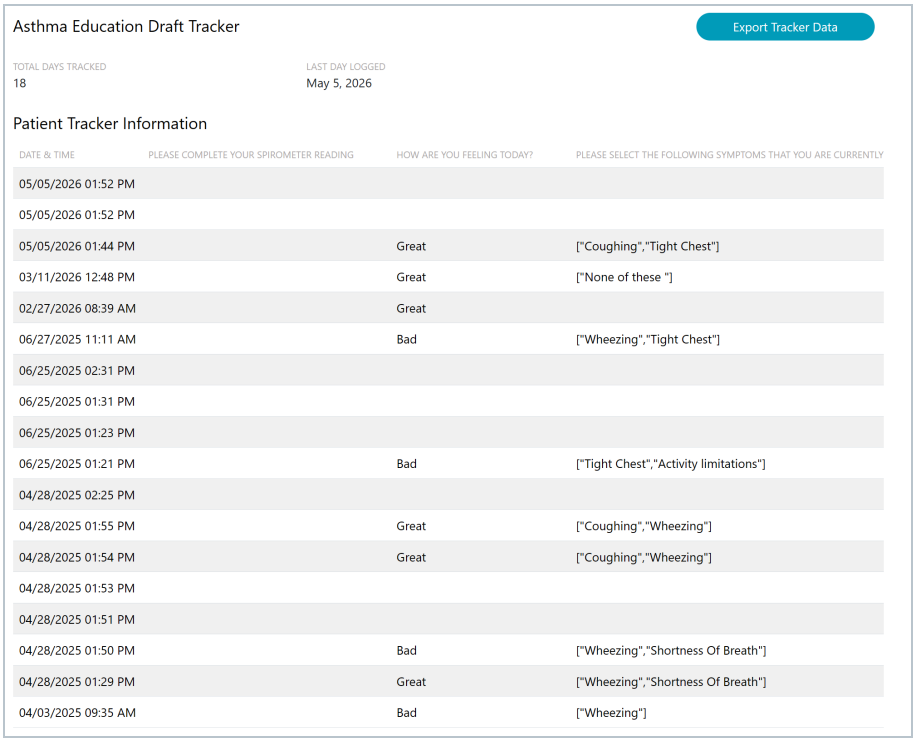
EARNED TRACKER BADGES

Badge Name	Date
TrackerBadge10	2026-03-11T16:48:59.546Z
TrackerBadge5	2025-05-01T00:16:23.561Z

Screenshot 1. The full patient record for one example account, laid out as a spreadsheet. Six separate tables are stacked down the page.

Screenshot 2. The clinician view in the web portal

This is what a clinician sees for a patient's asthma tracker in the SMART@Home web portal. The top shows two summary numbers: total days tracked (18) and last day logged (May 5, 2026). Below is a grid with one row per submission, newest first. Each column is one tracker question, with the full question as its heading.



Asthma Education Draft Tracker

Export Tracker Data

TOTAL DAYS TRACKED
18

LAST DAY LOGGED
May 5, 2026

Patient Tracker Information

DATE & TIME	PLEASE COMPLETE YOUR SPIROMETER READING	HOW ARE YOU FEELING TODAY?	PLEASE SELECT THE FOLLOWING SYMPTOMS THAT YOU ARE CURRENTLY
05/05/2026 01:52 PM			
05/05/2026 01:52 PM			
05/05/2026 01:44 PM		Great	["Coughing","Tight Chest"]
03/11/2026 12:48 PM		Great	["None of these "]
02/27/2026 08:39 AM		Great	
06/27/2025 11:11 AM		Bad	["Wheezing","Tight Chest"]
06/25/2025 02:31 PM			
06/25/2025 01:31 PM			
06/25/2025 01:23 PM			
06/25/2025 01:21 PM		Bad	["Tight Chest","Activity limitations"]
04/28/2025 02:25 PM			
04/28/2025 01:55 PM		Great	["Coughing","Wheezing"]
04/28/2025 01:54 PM		Great	["Coughing","Wheezing"]
04/28/2025 01:53 PM			
04/28/2025 01:51 PM			
04/28/2025 01:50 PM		Bad	["Wheezing","Shortness Of Breath"]
04/28/2025 01:29 PM		Great	["Wheezing","Shortness Of Breath"]
04/03/2025 09:35 AM		Bad	["Wheezing"]

Screenshot 2. The asthma tracker grid in the SMART@Home web portal, for the same example account.

Why it is hard to work with:

- **It is a raw log, not a summary.** To learn how the last month went, a clinician has to read every row and count by hand. There are no charts, no trends, no date filters and no flags for worrying entries.
- **The layout is fixed.** Column widths cannot be adjusted. Long question headings are cut off, for example "Please select the following symptoms that you are currently".
- **Spirometry is missing.** The "Please complete your spirometer reading" column is empty in every visible row, even though the device readings exist in the patient's record (Screenshot 1).
- **Many rows are blank.** Several rows have a date and time but no answers. The grid does not say whether the patient started and stopped an entry, completed a step shown elsewhere, or something else.
- **Repeated entries are not combined.** On April 28, 2025 there are six entries between 1:29 PM and 2:25 PM. Answers change within minutes, from "Bad" at 1:50 PM to "Great" at 1:54 PM. Nothing groups these into one day or picks out the answer that counts.
- **Symptoms are shown as code.** Answers appear with programming brackets and quotation marks, for example `["Coughing","Tight Chest"]`.

- **No context for answers.** A patient may report feeling "Great" while also reporting wheezing and shortness of breath. The grid gives the clinician no way to see that next to spirometry, medication use or the action plan zone.
- **Medications are not shown.** Whether the patient is taking the controller medication, one of the first things clinicians said they want to know, is not on this screen at all.
- **The summary numbers do not match the full record.** The page says the last day logged was May 5, 2026, but the full record in Screenshot 1 has medication doses and spirometer readings on June 24, 2026.

6. The hackathon challenge

Design a clinician view that turns these data into something a busy clinician can understand in a minute or two before a visit and can use with the patient during it. Start from the questions in section 4: how the disease is going, whether the patient is taking the medication, how often symptoms happen, and whether lung function is changing. Keep in mind that different specialists will want different views of the same data, and that the same view will need to work for trackers other than asthma.

Sources: SMART@Home Comprehensive Analysis (October 2026). POMIET clinician discovery reports (2022 and 2024). SMART@Home AI Research Handoff (October 2026). ClinicalTrials.gov NCT06159127.